An exact world audits its own experiments: what a small stochastic kernel can and cannot settle about learners

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Abstract

Exact synthetic worlds let us compare what a learned model acquires with what the data could teach it. We build one in which the observation channel is the controlled variable: a small stochastic operating-system kernel, observed through five nested interfaces, with exact posteriors and predictive laws for every window an observer can see. The uncertainty an interface leaves about the next event splits exactly into randomness that no observation could remove and hidden state the interface withholds. The mean randomness is the same on every interface by construction; the withheld part falls from about half of the uncertainty on the sparsest interface to about a third on the richest. We then use the same exact quantities to audit which learner experiments the world can support. The natural contrast between pure randomness and withheld state, at matched uncertainty, has about 1 % of the evaluated population on its withheld side, and 86–95 % of its random side is a single mechanism. Whether one prediction shows where its uncertainty comes from depends on the question and the precision. Among exact laws, telling pure randomness from substantial withheld state becomes ambiguous between separations of 0.01 and 0.02 in total variation, while the amount of randomness stays readable to 0.02 and becomes ambiguous by 0.05. For twelve trained transformers, measured by the prior mass of the classes whose exact law is at least as close to the model's output as the truth's, the kind of uncertainty meets a pre-declared readability rule in every cell and the amount of randomness in none. On the next event's kind, a comparison that separates an input rewrite from its training target suggests that the target contributes to the rewrite's gain; whether the input adds anything is unresolved at three seeds. An exact oracle makes such limitations measurable; it does not guarantee that the desired contrast exists.

1 Introduction

Exact worlds are popular for a reason. A transformer trained on Othello moves can be probed against the true board (Li et al., 2022), and transformers trained on sequences from hidden Markov models (Shai et al., 2024; Agarwal et al., 2025a) or on automata with known entropies (Someya et al., 2025) can be compared with exact Bayes quantities, because the information in the training data is known. A shortfall from the exact ceiling is then a property of the learner, not of unknown data.

Exactness removes one excuse, not all of them. Vafa et al. (2024) show that in exact finite-automaton worlds a standard diagnostic can still overstate how much of the world a model has recovered. We report a complementary set of limitations from one exact world, each measured with the world's own exact quantities: whether a planned contrast has support in the world, at what precision a question about a model's outputs becomes answerable, and whether an information-preserving intervention changes only what it was meant to change.

The world. Yupana is a small stochastic kernel (four threads, two CPUs, two locks, an asynchronous device) whose hidden state and transition law we own. The hidden system is held fixed while the *observation interface* varies: five nested interfaces each show one more field of every event record. For every window an observer can see, we compute the exact posterior over hidden state and the exact law of the next event.

Contributions.

1. **Exact measurements of what each interface makes knowable** (§3). Next-event uncertainty splits into randomness and withheld state; by construction, interfaces can change only the second on average. On the content interfaces, state uncertainty overstates the predictively relevant part by a factor of 3–5 while ranking fields identically. Where in the window the knowable information sits differs by interface, and the profiles cross.
2. **A support audit** (§4). For a planned contrast: the eligible mass, the matched mass, and the mechanism composition of the pure-randomness side, all computable from the exact tables before any training (we ran them retrospectively); and a precision sweep for an output-level identifiability question, placed beside the measured precision of trained models.
3. **A worked learner case** (§5), including a comparison that separates an intervention's input from its training target.

Motivation. The instrument was motivated by an impossibility result about verifying a model's epistemic state from its text alone (Mason & Anand, 2026): in a world where the truth is known, one can measure how much the observation channel withholds and how much a model's outputs reveal. §4.3 is the nearest this paper comes to that question, and it answers it only for one-step predictions in this world. It is an analogy to that result, not a test of it.

2 The instrument

2.1 The world

Yupana is a small stochastic operating-system kernel: four threads, two CPUs, two locks and one asynchronous device, with a request pool of four and a device queue of depth two. Each thread runs a fixed looping program, giving four contention roles. The kernel makes two kinds of random choice. The scheduler picks the next thread round-robin with probability $1 - \varepsilon$ and uniformly among runnable threads with probability ε; we report $\varepsilon = 1$ and $\varepsilon = \frac{1}{2}$. The device completes a pending request on a step with probability $1/3$, the request chosen uniformly among those pending. Every choice is labelled in the next record by its kind and actor, so the future stays random however much is observed.

An episode runs 40 records from reset. The observer sees a *window*: the last 8 records ending at an endpoint drawn uniformly from $\{2, 4, \ldots, 40\}$, with a RESET token when the window reaches back to the reset. This window law was fixed before any learner was trained.

2.2 Interfaces

Every record has five fields: KIND, ACTOR, OBJECT (which lock or device), RELATED (the thread woken, or the owner) and LINEAGE (the request identifier). Interface r0 shows only KIND, and each of r1–r4 shows one more field. Hidden fields are rendered as a placeholder, so every record is five tokens on every interface. Before any learner existed, each added field was shown to change some posterior or some finite-horizon predictive distribution.

Entity names carry no identity across episodes: they are drawn per episode from pools much larger than the entity counts (64 thread names for four threads, for example).

2.3 Exact quantities and their validation

Two observers appear. The *structural observer* sees structural identities; we use it only in §3.2. The *pattern observer* sees each entity renamed by order of first appearance, so that only the pattern of who recurs remains. Because names are redrawn every episode, it is the learner's comparison point, and every other quantity in this paper is for it, at position 7 (after seven records, predicting the eighth). The *ceiling* is the law-mass-weighted entropy of the exact predictive law given the pattern class: a Bayes predictive loss.

quantity	validation	coverage
structural window posteriors	the filtering recursion agrees with an independent path summation	exhaustive: 8,542,940 windows, all ten (interface, ε) cells
pattern-class predictive laws	agree exactly with an independent pattern filter	exhaustive at a small window law (episodes of 8 records, windows of 4, endpoint stride 2); at the frozen law, a random sample of 82–91 test windows per cell, all exactly equal
lag profile $\iota_m = H_m - H^*$	the KL route and the entropy route agree	every (interface, ε, m), to 3×10^{-14}
ceiling-holding rewrites	equal KIND marginal per class; equal multiset of record probabilities per class; an independent construction checked by brute force	exhaustive over classes
the R/G split (§3.1)	each interface's mean H^* reproduces the ceiling; $G \geq -10^{-15}$ per class (a mutual information, up to rounding)	every class, both ε

What none of this validates is the kernel. Every route calls the same kernel functions, so the gates check the computation *given* the kernel. The kernel rests on its tests, its reviews, and one property tested directly: from a reset, the r1 projection determines the structural state for every transition path up to eight records, at both ε. Appendix B specifies the kernel and the exact computation.

2.4 Learners

A decoder-only transformer (4 layers, width 128, 838,940 parameters) is trained from scratch on 2 M windows per cell with AdamW and a cosine schedule, for 5 or 15 epochs. At 15 epochs validation loss is still falling slowly, so we write "at 15 epochs" rather than "at convergence". The measurement is the KIND *shortfall* at position 7: the realised log-ratio between the exact law and the model's prediction, whose class mean estimates $KL(P^* \| q)$. On the content interfaces it runs at about 1–4 % of the ceiling. Here q is the model's distribution renormalised only to remove the end-of-window token, so probability placed on non-KIND tokens is penalised; §4.3 instead renormalises q over the KINDs, because it compares distributions over KIND. The trained model is the unit of inference: three seeds per cell, and inference per model or per matched seed pair. Appendix A lists which experiment was pre-registered, on which models and at what budget.

3 What each interface lets an observer know, exactly

Unless stated otherwise, the observer is the pattern observer, the position is after seven records, and the target is the next record's KIND. Position-7 quantities cover the windows that run at least seven records: 0.85 of window law mass.

3.1 Where the uncertainty comes from

The next-KIND entropy $H^*(c)$ of a class c splits exactly into two parts:

- $\mathbf{R(c) = \Sigma_s\ P(s \mid c)\ H(KIND \mid s)}$, the randomness an observer who knew the hidden state would still face;
- $\mathbf{G(c) = H^*(c) - R(c) = I(KIND; S \mid c)}$, the part due to state the interface withholds.

$\varepsilon = 1$	r0	r1	r2	r3	r4
H*, the ceiling	1.176	0.999	0.955	0.911	0.911
R, transition randomness	0.609	0.609	0.609	0.609	0.609
G, observation gap	0.567	0.391	0.346	0.302	0.302

Bits, law-mass means. At $\varepsilon = \frac{1}{2}$: R = 0.544 on every interface; G = 0.560, 0.329, 0.279, 0.240, 0.240.

R's mean is the same on every interface **by construction**: it is E[H(KIND | S)], which no observation changes. Per class, R does vary. The finding is in G: about half of the sparsest interface's mean next-KIND uncertainty is withheld state, and about a third of r3's (r4 agrees with r3 to within 10^{-5} bits on this target). On average, added fields act only on the withheld part; the mean randomness is beyond any interface's reach.

3.2 Knowing the state is not predicting the next event

The classical figure of merit for telemetry is identifiability: how much uncertainty about the hidden state the record leaves. Log placement work minimises a quantity of this kind (Zhao et al., 2017). For the structural observer at the window's last record, on the whole-record target:

$\varepsilon = 1$	r0	r1	r2	r3	r4
state entropy H(S | window)	3.947	0.601	0.528	0.405	0.390
next-record observation gap	0.442	0.150	0.135	0.106	0.103

Most state uncertainty is irrelevant to the next record: on r1, 0.60 bits of state uncertainty correspond to 0.15 bits of reducible next-record uncertainty. (The gap is a mutual information with the state, so it cannot exceed the state entropy; the size of the difference is the finding.) On the content interfaces, each further field (OBJECT, RELATED, LINEAGE) removes 3–5 times more of the first than of the second (4.7, 4.3, 4.4 at $\varepsilon = 1$; 4.3, 3.1, 4.0 at $\varepsilon = \frac{1}{2}$), and both measures rank those fields the same way. The first field, ACTOR, is not comparable, because it also changes the next-record target (r0's record is KIND alone); its ratio is 11.5 at $\varepsilon = 1$. Along this nested ladder, an identifiability objective would choose the same fields as a predictor's ceiling while overstating the relevant uncertainty by a fairly stable factor; that need not hold for unrestricted field selection. This table uses a different observer, position and target from §3.1, so the two should not be compared number by number.

3.3 Where in the window the knowable information sits

For a class c, let ι_m(c) be the information about the next KIND that lies more than m records back: the KL divergence from the full-window law to the law given only the last m records. Its law-mass mean is the interface's *lag profile*. It equals the order-m entropy minus the ceiling: a per-class, exact relative of the m-local entropy of Someya et al. (2025), with two differences of convention (our order m counts records and conditions on m of them, where theirs conditions on m − 1 symbols), computed for each interface of one fixed system.

The profiles cross. At lags 1–3, r3 and r4 (effectively tied; r4 is larger by under 10^{-5} bits) leave the most information beyond m records; from m = 4, r2 does ($\iota_4 = 0.282$ against r3's 0.259, $\varepsilon = 1$). A richer field both adds information and moves information that was already available closer to the point of use, and the ceiling registers only the first. The crossing is about 0.02 bits in one world; we report it as an illustration of why a profile is needed, not as a general effect.

3.4 Rewrites that hold the ceiling

A *window-level* rewrite may fill a masked slot with any function of the window's earlier records. It adds nothing an observer of the window lacked, so the ceiling is unchanged class by class, which three gates verify

exactly (equal KIND marginals per class; equal multisets of record probabilities per class; an independent construction checked by brute force on small worlds). We use one such rewrite in §5: the *relay*, which writes into each record the KIND of the same actor's previous record in the window.

4 Which contrasts can this world carry, and at what precision?

Exact ceilings tell a learner study what is knowable. They do not tell it whether the comparison it wants exists in the world: a contrast needs classes on both sides that match on everything but the quantity of interest. Nor do they tell it whether a question about a model's outputs can be answered at the precision those outputs have. The first check, and the exact half of the second, need only the exact class tables, so they could run before any model is trained; measuring a model's own precision needs the model. We ran them afterwards, on the learner studies of §5, and report what they would have shown.

4.1 A motivating error

Our first reading of the §5.1 results took a trained model's shortfall to be a *step*: roughly constant once a class has any far information ($\iota_4 > 0$), rather than rising with it. That reading came from fits that controlled the class's next-KIND entropy, and it does not survive. Classes with $\iota_4 = 0$ take only three next-KIND entropy values, so the entropy control was extrapolating far outside the support of one side of the contrast. In raw per-model bins, shortfall rises with ι_4 above $\iota_4 = 0.001$ in every model. Counting the entropy values on each side of $\iota_4 = 0$ would have shown the problem before the fit.

4.2 Randomness against withheld state

§3.1 splits each class's next-KIND uncertainty exactly into randomness R and withheld state G. The natural learner question is whether a model treats "the world is random here" differently from "the telemetry hides the state here". Its cleanest form compares classes of matched H* whose uncertainty is pure randomness (G = 0, H* > 0) with classes where much of it is withheld (G/H* ≥ 0.5), matched in 0.05-bit H* bins.

ε	interface	pure: eligible and matched (%)	of which the device coin (%)	high-gap: eligible (%)	high-gap: matched (%)	graded variation (%)
1	r1 / r2 / r3	14.6 / 15.6 / 16.7	92 / 91 / 86	39 / 35 / 28	1.4 / 1.2 / 0.9	21 / 24 / 19
$\frac{1}{2}$	r1 / r2 / r3	16.1 / 17.2 / 18.3	95 / 93 / 89	36 / 32 / 26	1.4 / 1.3 / 0.8	39 / 44 / 38

Shares of the position-7 population (law mass 0.85; r4 agrees with r3 at this precision). Pure: G = 0, H > 0; high-gap: G/H* ≥ 0.5; matched: sharing a bin floor(H*/0.05) with the other side. Every pure class is matched, and the matched mass lies in three shared bins. Device coin: a pure class whose H* is the entropy of a 1/3 coin (0.918 bits). Every such class has next-KIND law $\frac{1}{3}$ IO_COMPLETE and $\frac{2}{3}$ on a single other event: the kernel gives probability 1/3 to a completion whenever a request is pending and the remaining 2/3 to the scheduling step, which in these classes has one outcome (Appendix B). Graded variation: mass in bins where G/H* spans 0.30 or more between its mass-weighted 10th and 90th percentiles; restricting to bins with at least 0.5 % of mass and five classes gives 17–22 % (ε = 1) and 36–43 % (ε = $\frac{1}{2}$).*

Two things follow. The clean contrast is narrow: its high-gap side is about 1 % of the population, and 86–95 % of its pure-randomness side is one mechanism (a pending I/O completes with probability 1/3). A graded contrast has far more mass, but we have not audited how many mechanisms it spans and do not claim it is balanced. Small mass does not by itself make a contrast unusable; it says how much of the evaluation the contrast can draw on, and from what. The narrowness is a consequence of design: the kernel places randomness only in the scheduler and the device, to keep exact filtering inside its support budget.

4.3 Can one prediction say where the uncertainty comes from?

In general, the split between randomness and withheld state is not identifiable from a predictive distribution, even an exact one (Bergna et al., 2026); that result concerns unknown generative models. With the kernel known, structure can restore identification, so we ask the question inside the world for a stated family: the kernel, the interface and ε are known, and the observer is the pattern observer at position 7. Two senses of "where it comes from" matter, and they behave differently:

- the *kind*: pure randomness (G = 0 with H* > 0) against substantial withheld state (G ≥ 0.1 bits); classes in between have no kind. This differs from §4.2's high-gap side (G/H* ≥ 0.5), which asks for classes where withheld state dominates, for a matched learner contrast; here the question is only whether any substantial withheld state is present;
- the *amount of randomness* R.

Exact laws. Classes with identical exact next-KIND laws almost always share their split. The exceptions are below 0.05 % of position-7 mass on every interface, but real: on r3 at $\varepsilon = 1$, 24 reachable classes share the law $\frac{1}{3}$ ACQUIRE, $\frac{1}{3}$ IO_COMPLETE, $\frac{1}{3}$ RELEASE, with splits including (R, G) = (0.918, 0.667) and (1.585, 0).

Laws a small distance apart. Let δ be a total-variation separation between two classes' exact laws. Two laws can both lie within η of one prediction only if they are within 2η of each other, so δ corresponds to a prediction error of about $\delta/2$. For each class we ask whether some class within δ differs from it. These are existential statistics: one rare neighbour suffices.

δ	kind differs ($\varepsilon = 1$ / $\frac{1}{2}$, r1–r4)	R differs by ≥ 0.1 bits ($\varepsilon = 1$, r1 and r3)
0.01	0.1–0.2 % / 0.6–1.7 %	—
0.02	14–17 % / 17–19 %	1.8–2.4 %
0.05	16–18 % / 20–23 %	43–44 %

Shares of all position-7 mass. Opposite-sign R/G swaps of at least 0.05 bits each occur on about 3 % of mass at $\delta = 0.02$ and 18–21 % at $\delta = 0.05$ ($\varepsilon = 1$, r1 and r3).

The two senses fail differently over the separations evaluated. The kind becomes ambiguous between $\delta = 0.01$ and 0.02 and changes little from there to $\delta = 0.05$, the largest separation we evaluated. R stays pinned up to $\delta = 0.02$ and is ambiguous on 43–44 % of mass by 0.05. (G itself, the amount of withheld state, differs on 49–51 % of mass at $\delta = 0.02$, mostly because a small perturbation moves H*.)

Measured on trained models. We ran twelve trained models (r1 and r3, both ε, three seeds each) on their test windows. For each window, the *look-alike set* is the set of classes whose exact law is at least as close to the model's output as the true class's law is: $A(w) = \{c' : \mathrm{TV}(P^*_{c'}, q_w) \le \mathrm{TV}(P^*_c, q_w)\}$, where c is the window's class and q_w the model's output. The statistic is a(w) = Σ m(c′) over the c′ in A(w) that differ from c, divided by Σ m(c′) over all of A(w), with m a class's law mass. The set is calibrated by an oracle that knows the true class, so these are prior-mass shares of look-alikes, not error rates of any monitor. Before each run we declared a result *readable* if the mean of a(w) is below 0.05 and fewer than 10 % of windows have a(w) ≥ 0.10, *not readable* if the mean is at least 0.20 or at least 40 % of windows have a(w) ≥ 0.10, and otherwise neither; all three seeds of a cell must agree.

- The models' TV distance to the exact law: median 0.015–0.027, mean 0.027–0.041; 19–31 % of windows exceed 0.05.
- Look-alike mass of the other *kind*: 0.5–1.4 % on average; 1.3–3.9 % of windows have 10 % or more.
- Look-alike mass differing in R by 0.1 bits or more: 5–11 % on average; 10–23 % of windows have 10 % or more, higher at $\varepsilon = \frac{1}{2}$ and on r1 than on r3.

Three details of the kind measurement dilute its shares. Every class with G = 0, including deterministic ones, counted as pure; windows whose class has no kind (16–18 % of mass) score zero; and a window's own R and G were taken as the mean over classes sharing its exact law, which differs from the class's own value only on collision groups (below 0.05 % of mass). Deterministic classes and classes with no kind together carry 32–39 % of the mass, so at most 39 % of windows score zero by construction, and shares conditioned on the rest are at most 1/0.61 times those above: under 2.3 % on average and 6.4 % of windows.

We declared rules for calling each result readable or not before each run. The R measurement came first and fell between the rules in every cell. The kind measurement was declared after it, at a reviewer's prompting, and before it ran; it meets the "readable" rule in every cell and seed, with or without the correction above. For R we report the numbers only. Set against the exact sweep, the models' median error (0.015–0.027) allows separations of up to roughly 0.03–0.05, where the kind is already ambiguous on about a fifth of the mass existentially; that is only a compatibility bound. The look-alike measurement asks the more specific question at the models' actual outputs, and there the look-alike mass of the other kind stays small: ambiguous neighbours exist, but they are rare.

4.4 The audit as a procedure

The checks use only the exact class tables, which a study computing exact ceilings already has:

1. **State the estimand and the matching variables** (e.g. the shortfall difference between G = 0 and G/H* $\geq$ 0.5, matched on H* to 0.05 bits).
2. **Compute the eligible and matched mass on each side**, per interface and ε, as shares of the population the learner will be evaluated on.
3. **Report the mechanism composition** of each side, for example by grouping classes by their exact next-event law or by the kernel rule that produces it, with the dominant share.
4. **State the admissible family** for any identifiability claim (what a monitor is assumed to know), and test it inside the world, distinguishing the senses of the question.
5. **Sweep precision**: repeat any identifiability check over a range of separations; once models exist, place their measured output precision on the same axis.

The exact parts run on a workstation CPU from the existing tables. The model measurement needs one forward pass per model over its test windows; each twelve-model run took under a quarter of an hour of wall time as parallel CPU processes.

5 The learner case

The learners and the shortfall measure are those of §2.4. This is a measured case, not a general account of learners.

5.1 Shortfall is higher where far information is

Within each content interface, a pre-registered regression of per-window KIND shortfall d(w) on the class's far information, $\mathrm{d(w)} = \alpha + \beta\ \iota_4(\mathrm{c}) + \gamma\ (\iota_1(\mathrm{c}) - \iota_4(\mathrm{c})) + \lambda\ \mathrm{H^*(c)} + \mu\ \log_2 \mathrm{m(c)} + \mathrm{e(w)}$, with near information $\iota_1 - \iota_4$, ceiling entropy H* and log law mass m, gives $\beta > 0$ with $\mathrm{t} \geq 3$ in every run: 21 of 21 held-out runs at 5 epochs and 24 of 24 at 15 epochs. It survives errors clustered by class, and fixed effects for the last record's kind crossed with the number of distinct actors (lowest t 3.44), in all 45 of those evaluations (runs, not independent seeds). An exploratory check at about four times the parameters, in one cell, keeps the sign. The same analysis for the actor field passes its pre-registered test but is not robust to the fixed effects (9 of 24 runs keep $\mathrm{t} \geq 3$ at 15 epochs). The association is a correlation within cells: classes with more far information may also be rarer or harder in ways the controls miss. A pre-registered prediction that the coefficient would shrink no faster than overall shortfall with training failed in 8 of 8 cells. The size of the association at 15 epochs:

ε	interface	mean shortfall, $\iota_4 = 0$	mean shortfall, $\iota_4 > 0.4$	β with fixed effects (t)
1	r1 / r2 / r3 / r4	0.001–0.004	0.023–0.050	0.009–0.023 (3.6–8.1)
$\frac{1}{2}$	r1 / r2 / r3 / r4	0.001–0.004	0.035–0.061	0.013–0.024 (3.6–9.6)

Bits; ranges over the three seeds of each interface. β is bits of shortfall per bit of far information.

Baselines. A Witten–Bell count table over the ceiling's own classes, trained on the same windows, has 1.33–5.22 times the KIND shortfall of every transformer on every content interface. A parameter-matched GRU on the transformer's untuned recipe was pre-registered with a floor fixed before the models ran: each had to beat the count table before its association could be read. All eight failed it (KIND 0.045–0.076 bits against the table's 0.039–0.064), so their large, positive association statistics are uninformative under the declared rule.

5.2 A copy placed at the point of use: separating input from supervision

The relay rewrite (§3.4) holds the ceiling identical class by class while putting the same actor's previous KIND in each record. All three relay models had lower KIND shortfall than all three controls. A pre-registered prediction that the gain would scale with how much far information the relay moves closer failed. In the original training recipe the copy is both an input and a training target. Two pre-registered follow-ups separated the two by varying only the OBJECT slot. The five arms below are the cells that exist; no arm shows the copy while training on a placeholder, so the design is a set of controlled contrasts, not a full factorial (KIND and ACTOR shortfall at 15 epochs, seeds 1–3):

arm	OBJECT input	OBJECT target	KIND	ACTOR
r1	placeholder	placeholder	0.0157, 0.0146, 0.0158	0.0312, 0.0299, 0.0314
r1m	placeholder	none	0.0176, 0.0209, 0.0170	0.0346, 0.0438, 0.0409
r1Rs	placeholder	relay value	0.0125, 0.0141, 0.0142	0.0296, 0.0400, 0.0342
r1Rm	relay copy	none	0.0281, 0.0411, 0.0307	0.0328, 0.0363, 0.0345
r1R	relay copy	relay value	0.0130, 0.0095, 0.0125	0.0278, 0.0262, 0.0308

A contrast is *resolved* when every model of one arm lies beyond every model of the other in the pre-registered direction. With three seeds a side, exact paired sign-flip p-values cannot fall below 1/8, and none is conventionally significant. On KIND:

- **An unscored copy hurt** (r1Rm against r1m): showing the copy without training on it cost the learner, relative to a constant placeholder. This cannot distinguish an informative unscored copy from any varying field.
- **Changing the slot's target moved the shortfall** (r1Rs against r1): training the slot to produce the relay value, without ever showing it, lowered KIND shortfall in every seed, with a margin of 0.0005 bits between the closest models (unrounded values). Removing the slot from the loss altogether (r1m against r1) raised KIND shortfall by a comparable amount. KIND shortfall thus moves with any change to the OBJECT slot's loss, so r1Rs's gain cannot be credited to the content of the lookup rather than to having a nontrivial target in that slot. (Leaving OBJECT out of the loss also shrinks the loss denominator by about 24 %.)
- **Whether the copy adds to supervision is unresolved** (r1R against r1Rs).

On ACTOR, none of these contrasts resolves. The simple placement hypothesis, "a near copy helps a learned reader", finds no support here: the original gain does not establish an input-placement benefit. The training target contributes, and the input's increment remains open at three seeds.

For the audit of §4, this is a second kind of limitation an exact world leaves in place. Holding the ceiling fixed controls what is *knowable*. It does not control what the learner is *trained to produce*, and an information-preserving rewrite of the input can change the training targets with it.

6 Pitfalls we met that are not specific to exact worlds

Several of our errors are standard, and we only list them. Between-seed variance dominated paired comparisons, so the trained model is the unit (Bouthillier et al., 2021; Henderson et al., 2017). Far information measured on the original history and on the relay-rewritten history (§3.4) correlate at 0.79 over the paired test windows; in an exploratory follow-up with a declared reading, a comparison of which one tracks learner shortfall separated them, and in the one cell then checked the separation vanished once a reviewer added fixed effects for the last record's kind. Competing exact explanations should be compared under simple observable controls. Some output quantities are fixed by the training objective under stated conditions: a model whose log loss is stationary with respect to a global temperature, on the distribution and positions where it is evaluated, has zero mean one-step entropy-calibration gap (Braverman et al., 2019), and KL-regularised post-training has closed-form optima (Rafailov et al., 2023). Such quantities are reference predictions, not findings. And baselines need a quality floor fixed before they run (§5.1).

7 Related work

Exact testbeds. Othello-GPT (Li et al., 2022) and its successors; belief-state geometry in transformers trained on HMM data (Shai et al., 2024) and in pretrained LLMs given HMM data in context (Balcells et al., 2026); Bayesian "wind tunnels" (Agarwal et al., 2025a) and an entropy-aligned axis in production LLMs (Agarwal et al., 2025b). These comparisons vary the process (sampled transition and emission matrices) or the model; ours holds one process fixed and varies its observation interface. Vafa et al. (2024) evaluate world models in exact automaton worlds and show that common diagnostics mislead, and Vafa et al. (2025) probe foundation models for world models through their inductive bias, in settings including orbital mechanics. Our audit is a different check: support and precision, not recovery.

Locality and entropy convergence. Someya et al. (2025) relate learning to m-local entropy on natural and synthetic languages, measure learner excess loss against exact PFSA entropies, and use entropy-preserving transformations as interventions. Our lag profile is a per-class relative of their quantity (counted in records, conditioning on m of them), and our ceiling-preserving rewrites are close to their interventions; what differs is the fixed hidden system, class-by-class preservation, and the separation of input from supervision. Block-entropy convergence (Crutchfield & Feldman, 2003) is the classical form.

Separating kinds of uncertainty. The split of predictive uncertainty into aleatoric and epistemic parts is not identifiable from a predictive distribution in general (Bergna et al., 2026), whose decomposition is of predictive variance; ours is of entropy, conditioned on the hidden state. Joint or paired predictions can identify it (Osband et al., 2021; Johnson et al., 2024). The closest precedent is Ahdritz et al. (2024, Section 5.1): in a synthetic task whose questions carry an explicit bit marking each answer as fixed or resampled, one-step predictions are near 50/50 in both cases, and the difference appears only in how the model updates on a provided answer. Our §4.3 asks the one-step question in a world with no such cue, whose kernel is known, and finds that the answer depends on which sense of the question is asked.

Overlap. The support audit is positivity checking (Petersen et al., 2012; D'Amour et al., 2017) that can be applied before a learner study, using exact class masses.

Telemetry design. Log placement work optimises the identifiability of execution from logs (Zhao et al., 2017); §3.2 computes that kind of quantity beside the predictive one.

8 Limits

One world, with randomness placed by design in two mechanisms; one window law; one architecture (with an exploratory check at about four times the parameters in one cell); three seeds per cell; KIND at position

7. The identifiability results hold for the stated admissible family (known kernel, interface and ε, pattern observer, position 7). Learner precision was measured on twelve models on two interfaces. The kernel itself is validated only by its tests, its reviews and one directly tested property, and the pattern-class laws at the frozen law are validated on a sample of 82–91 test windows per cell rather than exhaustively. A reparameterised world, varying completion and dispatch probabilities with the posterior support unchanged, would test whether the audit's findings belong to this world's mechanisms.

Contribution statement

This research was conducted by successive instances of Claude (Anthropic) that owned the project, each continuing from the last through a timestamped Git history and written handoffs. The founding proposal was drafted with a ChatGPT (OpenAI) instance. Designs, pre-registrations and results were reviewed adversarially by separate Claude and Codex (OpenAI) instances; the repository records, for each, whether its review preceded the run or followed it. The reviews contributed substance as well as corrections, including the thesis sentence that closes the abstract and the collision example of §4.3. W. Anthony Mason originated the project, provided the hardware and the research framework (the *ayllu*, a collaboration organised around reciprocity between human and AI participants), took part as a collaborator, and posed questions that redirected the work; he is the author of record on behalf of the collaboration. Summary results and review records are in the repository; per-class and per-window data are regenerable from it and pinned by hash.

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A Which models carry which claim

experiment	cells	seeds per cell	budget	analysis
far-information association, KIND (held-out)	5 of the 8 r1–r4 cells	3; 6 in two cells	5 epochs	pre-registered before evaluation (21 runs)
far-information association, ACTOR	8 cells	36 runs: 21 held-out, 15 existing	5 epochs	pre-registered before any actor quantity
15-epoch ladder	8 r1–r4 cells, plus r0 at both ε	3	15 epochs	pre-registered before evaluation; three r1 $\varepsilon = 1$ models seen earlier
relay	r1 against r1R, $\varepsilon = 1$	3 per arm	5 and 15 epochs	pre-registered prediction (failed)
loss-masked arm	r1Rm	3	15 epochs	pre-registered

experiment	cells	seeds per cell	budget	analysis
relay follow-up contrasts	r1m, r1Rs	3 per arm	15 epochs	pre-registered contrasts
count baseline	10 cells	—	—	exploratory
GRU baseline	8 r1–r4 cells	1	15 epochs	pre-registered, with a quality floor
larger model	r1 $\varepsilon = 1$	3 (15 epochs), 1 (45 epochs)	15, 45 epochs	exploratory
learner precision (§4.3)	r1, r3 at both ε	3	15 epochs	two declared measurements; the second declared after the first was read

Training. AdamW (β 0.9, 0.95; weight decay 0.1), a 500-step linear warm-up to a learning rate of 10^{-3}, cosine decay to 10^{-4}, batch 256, gradient clipping at 1.0; the loss is the mean cross-entropy over scored target positions. At 15 epochs the last two epochs account for a median of 7.5 % of the decline in validation loss from epoch 5 (at most 17 %) across the 30 ladder runs. Test windows come from episodes disjoint from training.

Scoring. For the realised next record with KIND x, the KIND shortfall is $\log_2 \mathrm{P}^*(\mathrm{x} \mid \mathrm{class}) - \log_2 \tilde{q}(\mathrm{x})$, where $\tilde{q}$ is the model's softmax at the KIND position renormalised to remove only the end-of-window token. Class means estimate $\mathrm{KL}(\mathrm{P}^* \,\|\, \tilde{q})$; single windows can be negative. Subsets are always chosen by functions of the prefix, never by another realised field.

B The kernel and the exact computation

State. For each of the four threads: a program counter and a status (RUNNABLE; RUNNING; blocked on lock l; blocked on I/O request r; blocked on a full device queue; TERMINATED). Also: the set of threads on the two CPUs, each lock's owner and FIFO wait queue, the device queue of (thread, request) pairs (capacity two), and a round-robin cursor (absent from the dynamics at $\varepsilon = 1$). An episode starts with every thread RUNNABLE at program counter 0, no thread running, both locks free and the device queue empty.

Programs (each loops forever): thread 0 acquires lock 0, releases it, issues I/O; thread 1 acquires lock 0 then lock 1 and releases them in reverse order; thread 2 computes, then acquires and releases lock 0; thread 3 issues I/O, then acquires and releases lock 1.

One step emits one record. If a request is pending, the device completes one of the pending requests, chosen uniformly, with total probability 1/3 (record IO_COMPLETE, with the issuer as actor and the request as lineage); the issuer wakes. The remaining probability goes to the scheduling stage. If a CPU is free and a thread is runnable, a DISPATCH places a thread on a CPU for one instruction (uniform over runnable threads with probability ε, round-robin with probability $1 - \varepsilon$). Otherwise one running thread, chosen by the same rule, executes its next instruction: STEP for computation; ACQUIRE, or BLOCK with the owner as RELATED if the lock is held; RELEASE, with the lock handed directly to the head of its wait queue, who appears as RELATED; IO_ISSUE with a fresh request identifier as LINEAGE, or BLOCK on the device if its queue is full. If nothing can happen, the step is IDLE. A thread leaves the CPU after each instruction. The record kinds are thus DISPATCH, STEP, ACQUIRE, BLOCK, RELEASE, IO_ISSUE, IO_COMPLETE and IDLE, plus a RESET token at the start of a window that reaches back to the reset.

The exact computation. One recursion per (window law, ε) runs at the richest interface; coarser interfaces are projections. Its frontier maps (state, last-8-record window) to exact probability: at each tick every (state, window) is pushed through the kernel's transition distribution, and at each endpoint the frontier's mass is added, divided by the number of endpoints, to the joint law of (window, state). A *pattern class* at position

k is a window prefix of k records after renaming each entity type by order of first appearance, together with whether the window starts at a reset. Its law mass is the summed mass of its structural windows, and its next-record law is the posterior over (state, binding of pattern names to structural entities), pushed through one kernel step and re-keyed through the binding. Every probability is an exact rational; logarithms are taken in floating point at the end. Equality checks between independent routes are exact equality of rationals.

Size and cost at the frozen law: reachable states peak at 1,953 ($\varepsilon = 1$) and 4,173 ($\varepsilon = \frac{1}{2}$); the recursion's frontier peaks at 1,964,919 (state, window) pairs; one pass takes about nine minutes and about 5 GB on one workstation CPU. Pattern classes at position 7 number 5,540 (r0), 89,474 (r1), 116,624 (r2), 146,455 (r3) and 160,363 (r4).